
Structure-Based HMM Construction and Validation for Kunitz-Type Domain Detection

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1. ABSTRACT

Motivation: Kunitz-type domains are small, cysteine-rich protease inhibitor motifs found across microbes, invertebrates, and mammals, where they regulate key physiological processes such as blood coagulation, fibrinolysis, and inflammation. Reliably identifying remote Kunitz homologs in expanding sequence databases is challenging for sequence-based methods, since sequence identity between distantly related family members can fall below detectable thresholds even when the underlying 3D fold remains conserved. This work therefore builds a Hidden Markov Model (HMM) classifier from a structure-guided multiple alignment, to test whether this approach can improve remote homology detection for this domain family.

Results: 134 experimentally solved Kunitz-domain structures were retrieved from the PDB, filtered by length and annotation, clustered with MMseqs2 to reduce redundancy, and structurally aligned using mTM-align[1]; manual refinement yielded a final training set of 18 structures used to build a 58-match-state profile HMM with HMMER3. The model was validated against a curated, heavily imbalanced Swiss-Prot dataset (390 positive, 574,229 negative sequences) using 2-fold cross-validation, with the Matthews Correlation Coefficient (MCC) used as the primary metric due to class imbalance. The classifier achieved near-perfect discrimination ($MCC \geq 0.99$, $AUC = 1.0000$ in both folds) across a broad, stable range of E-value thresholds (10^{-4} to 10^{-8}). Structural validation of the small number of misclassified sequences (4 false negatives, 3 false positives) revealed two distinct failure modes: false negatives correspond to genuine, structurally confirmed Kunitz domains (TM-score = 0.600) that diverge at the sequence level beyond the profile's sensitivity, while false positives correspond to short, likely truncated protein fragments (TM-score = 0.355) that retain only a single conserved cysteine motif sufficient to trigger a locally significant HMM match.

Availability and Implementation: All scripts, the structure-based multiple alignment, the trained HMM model, and the supplementary materials used for this project are freely available at github.com/corbidaniela00/HMM_Kunitz_domain.

2. INTRODUCTION

The Kunitz-type domain represents a small, cysteine-rich structural motif of approximately 50 to 70 amino acid residues, found in diverse taxonomy groups including microbes, invertebrates and mammals. Structurally, the domain is characterized by a compact $\alpha + \beta$ fold consisting of two antiparallel β -sheets and one or two helical regions (Figure 1). This rigid architecture is structurally stabilized by three highly conserved disulfide bridges that follow a specific and invariant bonding pattern.[2] Kunitz domains play fundamental roles in regulating essential physiological cascades, including blood coagulation, fibrinolysis, and inflammation. In mammals, they frequently manifest as functional insertions within larger, complex proteins, such as the amyloid β -protein precursor (APP) associated with Alzheimer's disease pathology or the Tissue Factor Pathway Inhibitor (TFPI) involved in the coagulation cascade. Evolutionary adaptation has furthermore enabled venomous animals, such as

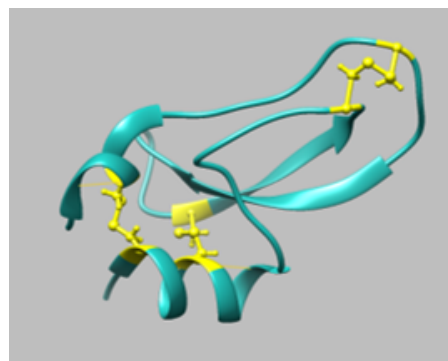


Figure 1. Tridimensional structure of 1LD5 (bovine pancreatic trypsin inhibitor) visualized with Chimera, disulfide bridges in yellow

snakes and spiders, to adapt these domains into Kunitz-type toxins (KTTs) that function as potent neurotoxins and are increasingly utilized as molecular scaffolds for protein engineering to design targeted therapeutic inhibitors.[2] Furthermore, novel Kunitz-type molecules like Amblyomin-X have demonstrated promising antitumor and antimetastatic activities by selectively triggering apoptosis in tumor cells, opening new frontiers for oncology treatments.[3] Given this biomedical significance, the accurate and comprehensive identification of

novel Kunitz domains within rapidly expanding genomic databases remains an important challenge. This project addresses this challenge by exploiting structural, rather than purely sequence-based, information to build a more sensitive Kunitz-domain classifier.

3. MATERIALS AND METHODS

3.1. Data sources

Experimentally determined protein structures annotated as Kunitz domain (PFAM: "PF00014"), with a polymer entity sequence length between 40 and 70 amino acids and a resolution $\leq 3.0 \text{ \AA}$ were retrieved from the RCSB Protein Data Bank (PDB). Results were retrieved in JSON format, in order to facilitate programmatic manipulation of entities, through a custom report containing the fields we were interested in, in particular

- Entry ID,
- PDB ID,
- Data Collection Resolution,
- Sequence,
- Polymer Entity Sequence Length,
- Annotation Identifier,
- Entity ID.

This query returned 134 structures satisfying the defined criteria.

3.2. Sequence Filtering and Clustering

Initial filtering was performed using the python script `filter_PDB_sequences.py`, to select only the chains of the sequences having the defined length and the defined annotation; this is necessary since the PDB query returns us all polymer entities of a structure having at least one entity satisfying the desired filters, including non-Kunitz co-crystallized chains. Filtered sequences were saved in a FASTA file that was processed by MMseqs2[4] in order to cluster sequences to address redundancy while preserving biological diversity. We applied a stringent 0.95 sequence identity threshold and a 0.85 coverage threshold. The increase of sequence identity in respect of the standard 0.8 follows the need to have more compact and homogeneous clusters, increasing consequently the total number in order to maintain the biological diversity, decreasing only the almost identical sequences. Raising the minimum alignment coverage to 0.85 guarantees that the alignment spans at least 85% of the sequence length, preventing "domain-trapping," where entirely different proteins are falsely clustered together simply because they share a single, short conserved motif

or binding domain. Using these constraints, we have ended with 20 clusters, from which a sequence per each was selected, the representative one, choosing the one with the lowest resolution through the script `select_one_seq_per_cluster.py`. The chosen sequences were then automatically inserted in `seed_alignmet_input.fasta` file.

3.3. Structural Alignment and Profile Visualization

The 20 representative PDB structures, taken automatically from the text file `lista.pdb.txt`, were downloaded dynamically from RCSB PDB using the file `download_structures_for_MSA.py`. Particular attention was applied in selecting only the chains we were interested in (the ones selected from the previous procedure of filtering). These structures were then subjected to Multiple Structure Alignment (MSA) using mTM-align software[1], which extends the TM-align pairwise algorithm to multiple structures. After visualizing the Multiple Sequence Alignment of the computed structure alignment with Aliview, we manually refined the alignment, reducing the redundancy, by removing two sequences (3FP7_J and 5NX1_D) almost identical to other two sequences (respectively 1G6X_A and 1AAP_A). It was then notice by manual check that they corresponded to the same UniProt code P00974 and P05067 respectively. The choice on which one retain was based on the coverage of all the 6 cysteine residues.

3.4. Model Construction

The Hidden Markov Model was built from the multiple sequence alignment derived from the multiple structure alignment, using the `hmmbuild` function from the HMMER3 package[5]. As shown in Figure 2, the resulting model comprises 58 match states, consistent with the expected length of a canonical Kunitz domain.

3.5. Validation Dataset

Positive and negative dataset of sequences, where for positive we intend annotated with the Kunitz domain (Pfam PF00014), were downloaded from the Swiss-Prot section of UniProt in order to deal only with curated entries. These sets are going to be used to validate the performance of our model in detecting proteins. To avoid data leakage, the 18 sequences used for model training were removed from the positive set after having mapped the PDB chain ID to UniProt Accession numbers using the UniProt ID Mapping Tool. After the matching we noticed there were some

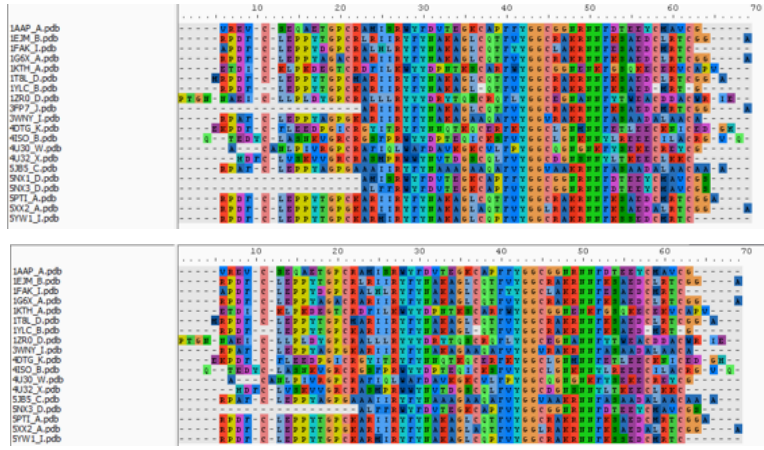


Figure 2. Multiple structure alignment of the 20 representative Kunitz-domain structures, before (top) and after (bottom) manual refinement.

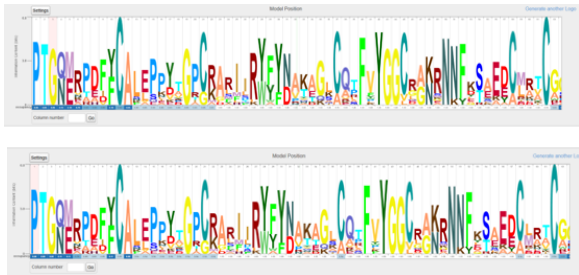


Figure 3. Sequence logos generated in Skylign from representative Kunitz-domain alignments used for HMM construction, before and after cleaning.

repetitions (multiple PDB sequence chain mapped to the same UniProt code) therefore have been removed from the positive only 8 sequences, ending in a positive set of 390 sequences and a negative one of 574,229 sequences. Since the dataset is heavily imbalanced the Matthews Correlation Coefficient was preferred over the accuracy as performance metric.

3.6. 2-Fold Cross Validation

Homology searches were performed using `hmmsearch` (HMMER3)[5] with the `--max` flag to disable heuristic filters and maximize sensitivity. To ensure cross-dataset comparability and reproducibility, the `-Z 1000` flag was applied to normalize E-values to a fixed search space size of 1,000 sequences. This standardization ensures that statistical significance reflects the biological quality of the match (Bit-score) rather than database volume. Furthermore, the 'best domain E-value' (column 8 of the tabular output) was used for classification, as it provides a more accurate metric for detecting Kunitz domains often embedded within larger multidomain proteins. The negative dataset was evaluated using `hmmsearch` against the Kunitz HMM model. Matches were extracted with their ID, corresponding E-values, while non-matching

sequences, lacking a valid E-value, were assigned a default value of 10, both groups were labeled with '0'. All negative sequences were then merged, randomly shuffled, and split into 2 balanced sets then combined with positive data (labeled as '1') and divided in 2 as well. the resulting datasets (`Kunitz_set.1.txt` and `Kunitz_set.2.txt`), each containing a mix of positive and negative examples, were used for 2-Fold cross validation for robust performance evaluation

4. RESULTS

Performance is evaluated using the confusion matrix, MCC vs threshold plot, and ROC curves computed using 2 python scripts `compute_performances.py` and `performance_analysys.py`.

4.1. Model Performance

Performance was assessed by evaluating the Matthews Correlation Coefficient (MCC) and Accuracy (Q2) across a wide range of E-value thresholds (10^{-1} to 10^{-15}), computed independently on each partition (`Kunitz_set.1`, `Kunitz_set.2`). For each fold of the 2-Fold Cross-Validation, the threshold maximizing MCC on the training partition is identified and subsequently applied to the corresponding held-out test set to produce the final confusion matrix. Both folds display a broad performance plateau between 10^{-4} AND 10^{-8} rather than a single sharp optimum (Figure 4). Within this plateau, MCC peaks at 10^{-5} on Set 1 (MCC = 0.9949), and at 10^{-4} for Set 2 (MCC = 0.9923, tied with several thresholds up to 10^{-8}). The two folds therefore do not select an identical optimal threshold, but the discrepancy is minor, reflecting that classifier performances are stable and not sensible to a specific threshold. The resulting Confusion Matrices (Figure 5) show three false negatives when training on Set 1 (threshold 10^{-5}) and three false positives plus

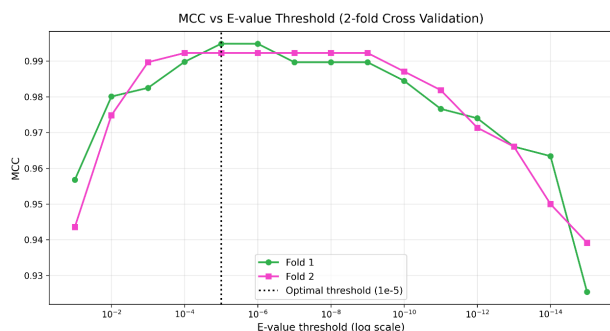


Figure 4. MCC (Matthews Correlation Coefficient) as a function of the E-value threshold, on a log scale, separately for the two folds of your 2-fold cross-validation

one false negative when testing on Set 2 (threshold 10^{-4}). These cases are examined in detail in sections 4.2 and 4.3.

Receiver Operating Characteristic (ROC) curves for both folds (Figure 6), using the negative log-transformed E-value as the discriminative score, achieve an AUC of 1.0000 with the operating point corresponding to the cross-validated optimal threshold from the opposite fold lying at the curve’s corner (FPR ≈ 0 , TPR ≈ 1), confirming that the classifier achieves near-perfect separation between Kunitz and non-Kunitz sequences.

4.2. False Negative Analysis

The four false negatives (Table 1) all carry a valid Prosite PRU00031 annotation, are AlphaFold-predicted models (no experimental structure available) and are noticeably longer than the canonical ≈ 58 -residue Kunitz domain, consistent with the domain being embedded within a larger multi-domain, full-length protein context. This is a plausible explanation for reduced sensitivity: the profile HMM must detect the domain against a longer, more heterogeneous sequence background, which can dilute the local signal and inflate the E-value even when the domain itself is genuinely present. To test whether these sequences adopt the Kunitz fold despite being missed by the HMM, a multiple structural alignment (MSTA) was performed between their AlphaFold-predicted models[6] and a canonical Kunitz domain reference structure (1AAP_A, amyloid precursor protein). The overall TM-score of 0.600, above the commonly accepted threshold of 0.5 used to indicate that two structures share the same fold, confirms that all four false negative proteins adopt a Kunitz-like fold despite their substantial divergence at the sequence level. The average aligned length (Lali = 63.7 residues) and low RMSD (1.387 Å) confirm close geometric agreement between the

superimposed structures. However, restricting the comparison to the common structural core, the subset of positions consistently superimposable across all five structures simultaneously (Lcore = 45 residues), the core TM-score (ccTM-score = 0.490, just below of the 0.5 similarity threshold), while the core RMSD (ccRMSD = 1.051 Å) remains low. This suggests that the structural divergence between the four false negatives and the reference domain is concentrated in a subset of positions outside this strict core, most likely corresponding to loop regions.

This confirms that the source of the model’s reduced sensitivity lies in sequence-level divergence not captured by the profile, despite the HMM being built from a structure-guided alignment, rather than in an absence of the underlying fold itself. All four E-values (0.00031–0.062) fall well above the 10^{-5} – 10^{-4} threshold range, with no borderline cases close to the decision boundary. This indicates that these sequences are missed with a comfortable margin rather than being marginal cases sensitive to small threshold shifts, and is consistent with genuine evolutionary divergence between these distantly related family members and the training set used to build the profile HMM, which likely skews toward vertebrate/BPTI-like sequences. Improving recall on this class of true positive would likely require enriching the training alignment with more phylogenetically diverse Kunitz domain representatives, rather than adjusting the classification threshold.

4.3. False Positive Analysis

The three false positives (Table 2) all carry a valid Prosite PRU00031 annotation, indicating that they belong to the Kunitz/BPTI family rather than representing spurious hits. Their misclassification is therefore not attributable to random noise in the model, but to a specific structural feature shared by all three entries. All three sequences are AlphaFold-predicted models, markedly shorter (≈ 20 – 30 residues) than the canonical ≈ 58 -residue domain and present only 2 of the 6 expected cysteines. In each case the two cysteines are annotated as forming a disulfide bond, but the identity of the partner cysteine is not resolved in the UniProt entry, a strong indication that the deposited sequence represents a partial or truncated fragment of the full-length domain, rather than the complete structural unit. A multiple structural alignment against the same reference structure (1AAP_A) supports this interpretation clearly. In contrast to the false negative set, the average length of only 31.75 residues,

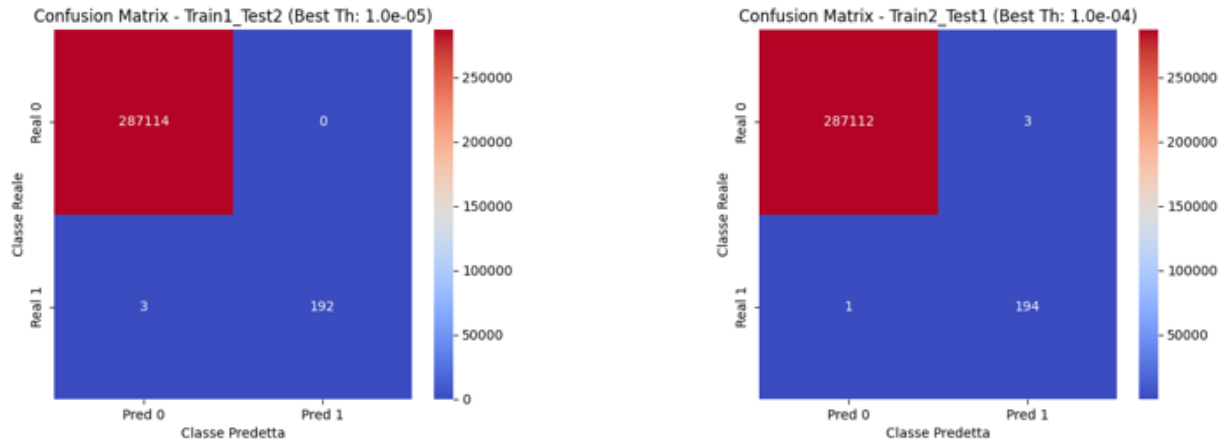


Figure 5. (1) confusion matrix considering Kunitz_set.1 as training set and Kunitz_set.2 as Testing test having obtained as best threshold 10^{-5} (2) confusion matrix considering Kunitz_set.2 as training set and Kunitz_set.1 as Testing set having obtained as best threshold 10^{-4}

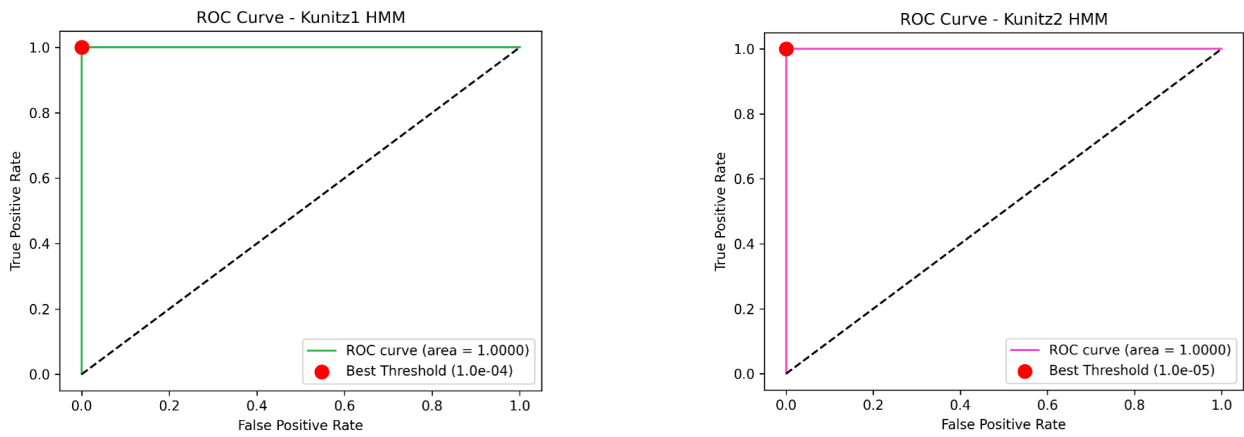


Figure 6. ROC curves for both folds of the 2-fold cross-validation.

Table 1. False Negative Analysis

UniProt ID	Protein	E-value	Annotation
O62247	bli-5	0.015	PRU00031 (Prosite)
D3GGZ8	bli-5	0.062	PRU00031 (Prosite)
A0A1Q1NL17	HA11	0.0028	PRU00031 (Prosite)
Q8WPG5	Thrombin inhibitor savignin	0.00031	PRU00031 (Prosite)

Table 2. False Positive Analysis

UniProt ID	Protein	E-value	Annotation
P83605	BmTI-4	2×10^{-5}	PRU00031 (Prosite)
P84555	RsTIQ7	1.8×10^{-6}	PRU00031 (Prosite)
P0DJ63	Kunitz-type serine protease inhibitor	2.2×10^{-5}	PRU00031 (Prosite)

roughly half that of a canonical Kunitz domain and the overall TM-score (0.355) show that these structures do not share the canonical Kunitz fold in any statistically meaningful way. The core alignment metrics reinforce this conclusion even more clearly: the common structural core comprises only 7 residues ($L_{\text{core}} = 7$), compared to 45 residues found in the false negative analysis, and the corresponding core TM-score (ccTM-score = 0.211) is drastically lower. Visual inspection of the structural alignment shows that the only consistently superimposable region across all three false positives corresponds to a short stretch centered on a single conserved cysteine and its immediately flanking residues, while the remainder of each sequence shows no consistent structural correspondence. This suggests a specific failure mode of the profile HMM: the model assigns high information content to a small number of highly conserved local motifs surrounding the cysteine residues. When a short fragment happens to retain one such locally conserved motif, it can achieve a statistically significant E-value (2×10^{-5} , 1.8×10^{-6} , and 2.2×10^{-5}) even though it covers only a small fraction of the full domain. This reflects a known limitation of E-value-based classification: local conservation at a few high-weight positions can outweigh the absence of global domain coverage.

5. CONCLUSION

This project develops and validates an HMM classifier for Kunitz-domain detection, built from a structure-guided multiple alignment of 18 experimentally solved PDB structures. The model achieves near-perfect discrimination on a large, imbalanced Swiss-Prot validation set ($\text{MCC} \geq 0.99$, $\text{AUC} = 1.0000$), with stable performance across several orders of magnitude of E-value thresholds rather than depending on a single fragile cutoff. The structural validation of the small number of misclassified sequences proves to be the most informative part of this evaluation, revealing two distinct error sources: false negatives are genuine, structurally confirmed Kunitz domains, missed due to sequence divergence beyond the profile's sensitivity, likely reflecting the training set's bias toward vertebrate/BPTI-like sequences, while false positives are short, truncated protein fragments that satisfy the E-value threshold by chance, through a single conserved cysteine motif, without possessing the complete fold. This shows that building the HMM from a structural alignment did not eliminate classification errors, but made them explainable rather than arbitrary. Two concrete directions for

improvement follow: (i) enriching the training alignment with more phylogenetically diverse Kunitz representatives to reduce false negatives, and (ii) complementing the E-value criterion with a minimum domain-coverage filter to exclude partial fragments and reduce false positives.

6. REFERENCES

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